RESEARCH SOFTWARE - VERSION 2.0.0

MohandesYar AI 2.0: An Offline-First Persian PWA for Civil Engineering Field Documentation, Evidence Integrity, and Reporting

A local-first workflow for project dossiers, field evidence, geolocation metadata, backup and restore, and multi-page Persian A4 reports.

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Repository: github.com/y0bahrambeigi/bhb

Abstract

MohandesYar AI 2.0 is an offline-first progressive web application designed for Persian right-to-left field documentation in civil engineering projects. The public edition supports multiple project dossiers, storage of original image and video evidence in the browser, optional geolocation capture, SHA-256 integrity metadata, project backup and restoration, and generation of multi-page Persian A4 reports. The architecture intentionally follows a local-first model: project data and evidence remain in IndexedDB on the user's device and are not transmitted to the public hosting service. Release validation combines static contract checks with browser-based end-to-end tests covering persistence, evidence hashing, backup and restore, report rendering, service-worker updates, IndexedDB retention, and offline relaunch. The validated scenario retained six images and one video, recorded geolocation metadata for all seven items, restored evidence from backup, generated an eight-page report, and preserved data through a service-worker update and an offline restart. The software is intended as a practical documentation aid rather than an institutional record system. It does not provide server-side identity, role-based access control, digital signatures, organizational audit trails, or automatic submission to authorities. This report documents the software scope, architecture, validation evidence, limitations, and reproducibility information for version 2.0.0.

Keywords: civil engineering; field documentation; progressive web application; offline-first; Persian RTL; IndexedDB; SHA-256; geolocation; evidence integrity; technical reporting

Record	Value
Software	MohandesYar AI
Version	2.0.0
Author	Yousef Bahrambeigi
Release date	25 August 2026
DOI status	Pending public Zenodo release
Live edition	https://y0bahrambeigi.github.io/bhb/mohandesyar-ai/
Source code	https://github.com/y0bahrambeigi/bhb/tree/main/mohandesyar-ai

1. Motivation and scope

Civil engineering inspections routinely produce heterogeneous evidence: project identifiers, visit notes, photographs, short videos, geolocation data, findings, and recommended actions. When these items are distributed across camera rolls, messaging applications, and handwritten notes, traceability and report preparation become difficult. MohandesYar AI 2.0 consolidates this workflow in a browser-based application that can be installed on mobile and desktop devices and can continue operating after network loss.

The version 2.0.0 public edition is intentionally single-device and local-first. It supports individual professional use and demonstrative field workflows. It is not presented as a replacement for organizational document management, legally binding inspection systems, or regulatory submission portals.

2. System architecture

Layer	Implementation	Purpose
Interface	Static HTML, CSS and ES modules	Persian RTL dashboard, project forms, evidence controls and reporting
Persistence	IndexedDB	Local storage of project records, evidence blobs and application metadata
Integrity	Web Crypto SHA-256	Content fingerprint for each stored evidence file
Location	Browser Geolocation API	Optional latitude, longitude and accuracy capture after user permission
Offline	Service worker and scoped cache	Installation, cached routes, update isolation and offline relaunch
Reporting	Dynamic HTML and print CSS	Multi-page A4 Persian report with evidence details and watermark

3. Field evidence and data integrity

Each evidence record links the original browser file blob to the active project and stores the file name, media type, byte size, capture time, optional note, optional geolocation coordinates, reported location accuracy, and a SHA-256 content hash. The hash supports later equality checks and makes unintended file substitution detectable. During backup restoration, version 2.0.1 hardening recomputes evidence hashes and rejects mismatches rather than trusting user-supplied metadata.

Integrity metadata is not a digital signature. A hash demonstrates content identity only when the trusted reference value and the file are managed correctly. The public edition does not authenticate the operator, establish legal chain of custody, timestamp evidence through a trusted authority, or prevent a privileged device user from changing local records.

4. Privacy, backup and restoration

- Project data and media remain in the browser origin's IndexedDB storage and are not uploaded by the public GitHub Pages edition.
- Geolocation is read only when the user selects location capture and grants browser permission.
- Backup export serializes project data, evidence metadata and evidence payloads into a local JSON file.
- Restore supports merge and full replacement modes; replacement requires explicit confirmation.
- Clearing site data, uninstalling the browser, resetting the device or exhausting browser quota may remove local evidence, so periodic external backups remain essential.

5. Persian reporting workflow

The reporting module reads the active project and its stored evidence directly from IndexedDB. The report contains project details, inspection description, findings, recommended actions, evidence images, file metadata, geolocation data, hashes, and a visible status watermark. Print CSS targets A4 output, preserves RTL layout, reduces clipping through break controls, and waits for evidence images to load before enabling printing. Video evidence is represented by metadata and its hash; video payloads are not embedded in the PDF.

6. Release validation

The release pipeline combines static validation with an automated Chromium scenario and manual device checks. Static tests verify JavaScript syntax, JSON manifests, public routes, Persian RTL declarations, IndexedDB use, SHA-256 hashing, optional geolocation, print layout contracts, service-worker cache isolation, and stable database identity. Browser QA exercises the complete user workflow.

Validation target	Observed result	Status
Evidence persistence	Six images and one video stored and reopened from IndexedDB	PASS
Geolocation	Coordinates recorded for all seven test items	PASS
Content hashes	SHA-256 retained for all stored evidence	PASS
Backup and restore	Payload exported, test data removed, and all evidence restored	PASS
Persian PDF	Eight-page RTL report with six non-empty images and watermark	PASS
Update retention	Service worker updated while project and evidence remained	PASS
Offline relaunch	Application closed and reopened offline with seven items available	PASS
Physical devices	Android Chrome and iPhone Safari installation paths reported as passed	PASS*

* Physical-device results were recorded in the release notes. Device model, operating-system version and browser build were not supplied, which limits independent reproducibility of that subset.

7. Operational limitations and risk boundary

- No user account, server-side identity, organizational role model or centralized access control.
- No cloud synchronization, server backup, institutional retention policy or administrator recovery.
- No trusted timestamp, digital signature, immutable audit trail or legal chain-of-custody service.
- No automatic transmission to municipalities, engineering organizations, employers or clients.
- Browser storage quotas and device lifecycle remain outside the application's control.
- The term AI is part of the product name; version 2.0.0 does not transmit data to an external AI inference service.

8. Reproducibility and availability

The application source is available in the public BHB repository. The release verification command runs syntax checks and static publication-contract tests; the browser QA command executes the end-to-end scenario. The live GitHub Pages deployment provides the installable public edition. Users should archive the cited version and commit; an archival DOI will be added after public activation.

Purpose	Command or address
Static verification	cd mohandesyar-ai && npm run verify
Browser QA	cd mohandesyar-ai && npm run qa:browser
Repository	https://github.com/y0bahrambeigi/bhb
Live application	https://y0bahrambeigi.github.io/bhb/mohandesyar-ai/
DOI status	Pending public Zenodo release

9. Recommended citation

Bahrambeigi, Y. (2026). MohandesYar AI 2.0: An Offline-First Persian PWA for Civil Engineering Field Documentation, Evidence Integrity, and Reporting (Version 2.0.0) [Technical report]. BHB Smart Structures Lab. Report MYAI-TR-2026-02.

Indexing metadata

Field	Value
citation_title	MohandesYar AI 2.0: An Offline-First Persian PWA for Civil Engineering Field Documentation, Evidence Integrity, and Reporting
citation_author	Yousef Bahrambeigi
citation_publication_date	2026/08/25
resource type	Technical report
version	2.0.0
DOI status	Pending public Zenodo release

References

[1] Bahrambeigi, Y. MohandesYar AI source repository and release documentation. GitHub, 2026.

[2] Bahrambeigi, Y. MohandesYar AI 2.0 technical report. BHB Smart Structures Lab, MYAI-TR-2026-02, 2026.

[3] National Institute of Standards and Technology. FIPS PUB 180-4: Secure Hash Standard (SHS). 2015. https://doi.org/10.6028/NIST.FIPS.180-4

[4] World Wide Web Consortium. Indexed Database API 3.0. https://www.w3.org/TR/IndexedDB-3/

[5] World Wide Web Consortium. Geolocation API. https://www.w3.org/TR/geolocation/

Disclosure: This report documents the public software release and its recorded validation evidence. It does not certify regulatory compliance or confer legal validity on reports produced by the application.